Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2025 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

31 working units   106 marks

Selected from Linear Higher Papers 1H and 2H

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Elite IGCSE Mathematics • eliteigcse.com

L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each page contains one main demand or one small inseparable chain. Source panels are followed by adaptive working space; writable diagrams count as working space.

Unit ownership rule: a question belongs to Unit 2 when its assessed, new or terminal statement is Unit 2, even when the working uses Unit 1 prerequisites. For example, the region-of-inequalities question remains Unit 2 although drawing and reading straight lines is supporting Unit 1 knowledge.

Contents

1 The table gives information about the distances 100 adults travel to work.

Distance (d km)	**Frequency**
$0 < d \leqslant 5$	26
$5 < d \leqslant 10$	40
$10 < d \leqslant 15$	16
$15 < d \leqslant 20$	10
$20 < d \leqslant 25$	8

(a) Write down the modal class.

....

(1)

(b) Work out an estimate for the mean distance.

.... km

(4)

WORKING SPACE

2 Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

(a) Work out the percentage profit Anna makes.
Show your working clearly.

....%

(4)

WORKING SPACE

3

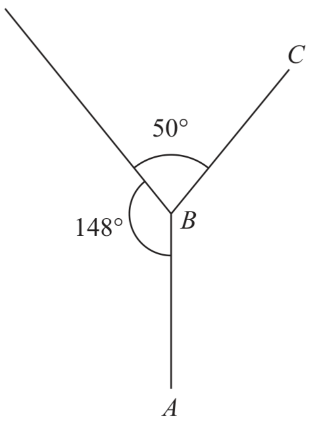


Diagram **NOT** accurately drawn

AB and *BC* are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

n = ...

(Total for Question 3 is 4 marks)

WORKING SPACE

5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.

Work out the normal price of the watch.

.... euros

(Total for Question 5 is 3 marks)

WORKING SPACE

7 Here is a solid cylinder.

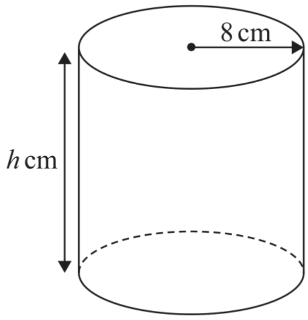


Diagram **NOT** accurately drawn

The radius of the cylinder is 8 cm
The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h
Give your answer correct to one decimal place.

h = ...

(Total for Question 7 is 3 marks)

WORKING SPACE

8 (a) Write 520 million in standard form.

....

(1)

(b) Write 8.79×10^{-5} as an ordinary number.

....

(1)

WORKING SPACE

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

....
(2)

WORKING SPACE

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%
In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by x%

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

x = ...

(Total for Question 13 is 3 marks)

WORKING SPACE

16 $f(x) = 9 - \sqrt{x}$ where $x \geqslant 0$

$g(x) = 4x^2$

(a) Find f(9)

....

(1)

WORKING SPACE

16 $f(x) = 9 - \sqrt{x}$ where $x \geqslant 0$

$g(x) = 4x^2$

(b) Solve $fg(x) < 0$

Show clear algebraic working.

....

(3)

WORKING SPACE

18 y is inversely proportional to x^3

$y = 4$ when $x = 3$

Work out the value of x when $y = 864$

$x =$ ....

(Total for Question 18 is 4 marks)

WORKING SPACE

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.......................... , ...)
(1)

(ii) $y = f(2x)$

(.......................... , ...)
(1)

(Total for Question 19 is 2 marks)

WORKING SPACE

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$
$$y = x + 2$$

Show clear algebraic working.

....

(Total for Question 22 is 5 marks)

WORKING SPACE

23

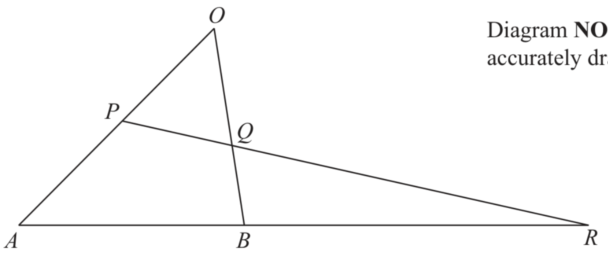


Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of OA
Q is a point on OB

ABR and PQR are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\qquad \overrightarrow{OB} = 8\mathbf{b}$

(a) Express $\overrightarrow{AB}$ in terms of **a** and **b**

....
(1)

WORKING SPACE

Continued

4MA1-1H Q23

UNIT 2

23

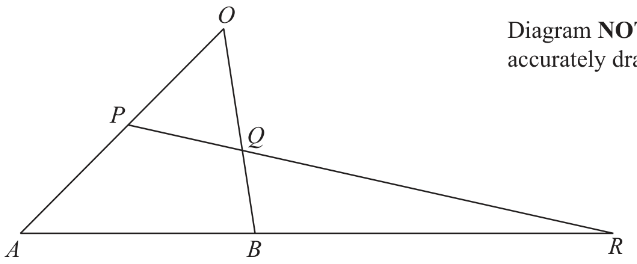


Diagram **NOT** accurately drawn

OAB is a triangle.
P is the midpoint of OA
Q is a point on OB

ABR and PQR are straight lines.

$\overrightarrow{OA} = 12\mathbf{a}$ $\qquad$ $\overrightarrow{OB} = 8\mathbf{b}$

$AB : BR = 1 : 2$ $\qquad$ $\overrightarrow{OQ} = n\mathbf{b}$

(b) Use a vector method to find the value of n

$n =$ ....

(4)

WORKING SPACE

24 An arithmetic series has 30 terms.

The first term is a
The common difference is d

The 20th term is 123
The sum of the 30 terms is 2880

Work out the value of a and the value of d
Show clear algebraic working.

$a =$ ....

$d =$ ....

(Total for Question 24 is 5 marks)

WORKING SPACE

1 Rio has six cards.
He writes a number on each card so that

the range of the numbers is 10
the median of the numbers is 7.5
the mode of the numbers is 6

Rio arranges the cards so that the numbers are in order of size.

		6		10	14

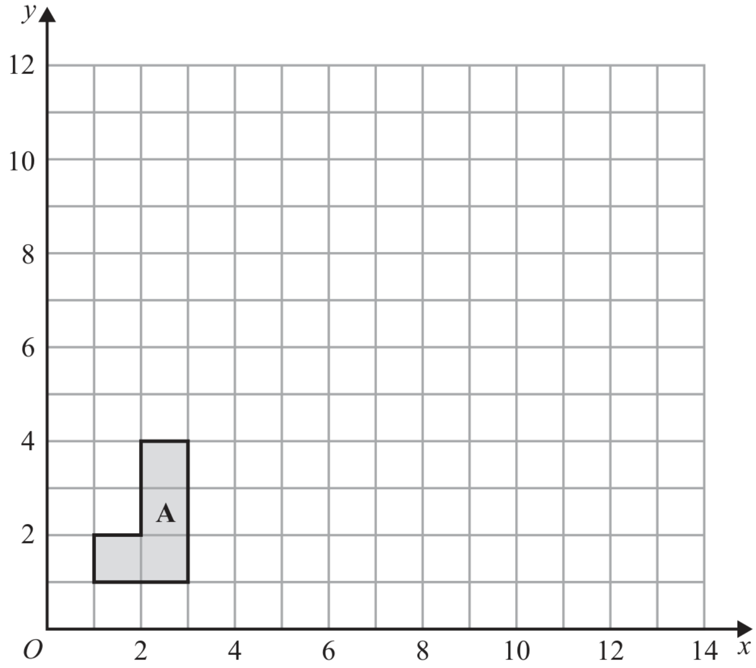
Three of the numbers are hidden.

Complete the cards above to show the three numbers that are hidden.

(Total for Question 1 is 3 marks)

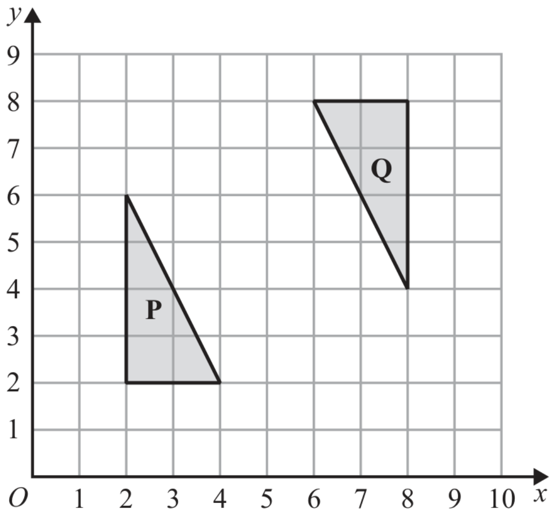
WORKING SPACE

2



(a) On the grid, enlarge shape **A** with scale factor 3 and centre (0, 1)

(2)



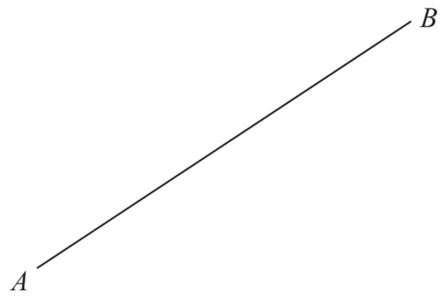
(b) Describe fully the single transformation that maps triangle **P** onto triangle **Q**

....

....

(3)

4 Using ruler and compasses only, construct the perpendicular bisector of the line AB
Show all your construction lines.



(Total for Question 4 is 2 marks)

5 The diagram shows two similar quadrilaterals, *ABCD* and *EFGH*

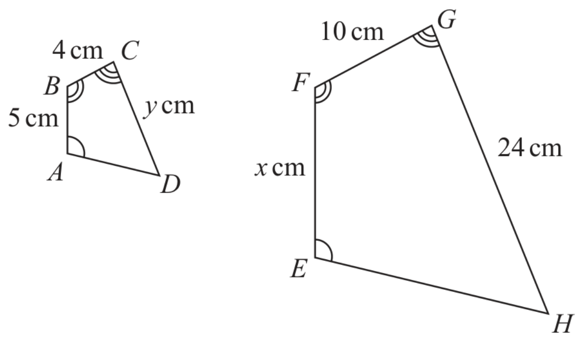


Diagram **NOT** accurately drawn

$AB = 5$ cm $BC = 4$ cm $CD = y$ cm

$EF = x$ cm $FG = 10$ cm $GH = 24$ cm

(a) Work out the value of x

$x =$ ....

(2)

WORKING SPACE

Continued
4MA1-2H Q5

UNIT 2

5 The diagram shows two similar quadrilaterals, *ABCD* and *EFGH*

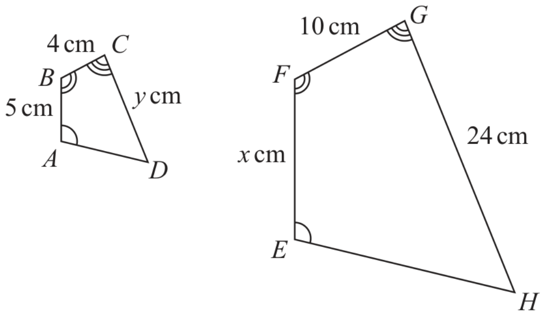


Diagram **NOT** accurately drawn

$AB = 5\text{ cm}$ $BC = 4\text{ cm}$ $CD = y\text{ cm}$

$EF = x\text{ cm}$ $FG = 10\text{ cm}$ $GH = 24\text{ cm}$

(b) Work out the value of y

$y =$ ....

(2)

WORKING SPACE

6 Pau, Sam and Tia share £240 in the ratios $3:4:5$

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

.... : .... : ....

(Total for Question 6 is 4 marks)

WORKING SPACE

7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

.... Swiss francs

(Total for Question 7 is 3 marks)

WORKING SPACE

8 Solve the simultaneous equations

$$3x + 5y = 8$$
$$4x + y = -3.5$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 8 is 3 marks)

WORKING SPACE

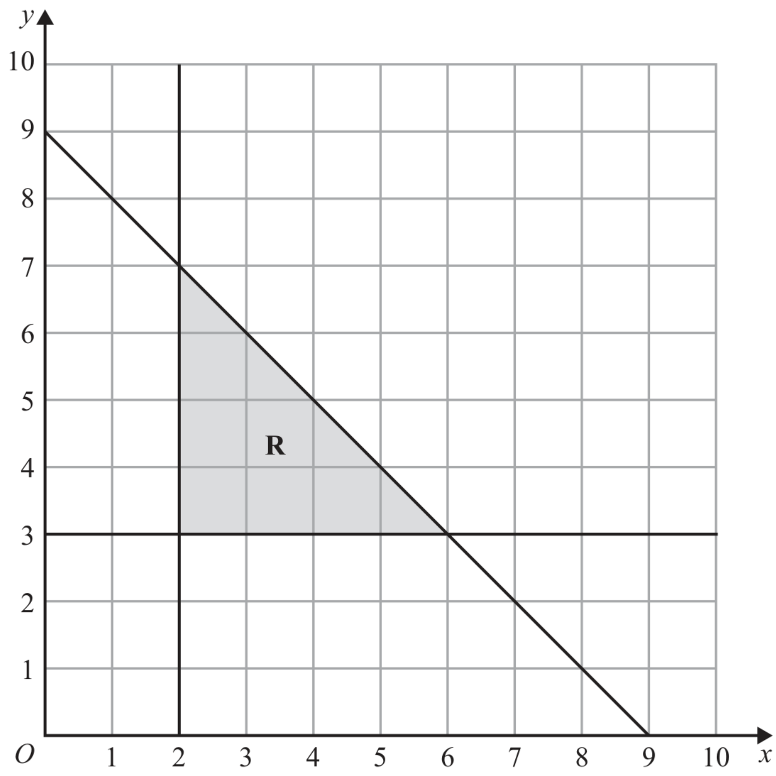
9 (a) Solve the inequality $7 - 3t < 2t + 15$

....

(2)

WORKING SPACE

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

....

....

....

(3)

13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	**Frequency**
$0 < t \leqslant 10$	8
$10 < t \leqslant 20$	13
$20 < t \leqslant 30$	12
$30 < t \leqslant 40$	17
$40 < t \leqslant 50$	7
$50 < t \leqslant 60$	3

(a) Complete the cumulative frequency table.

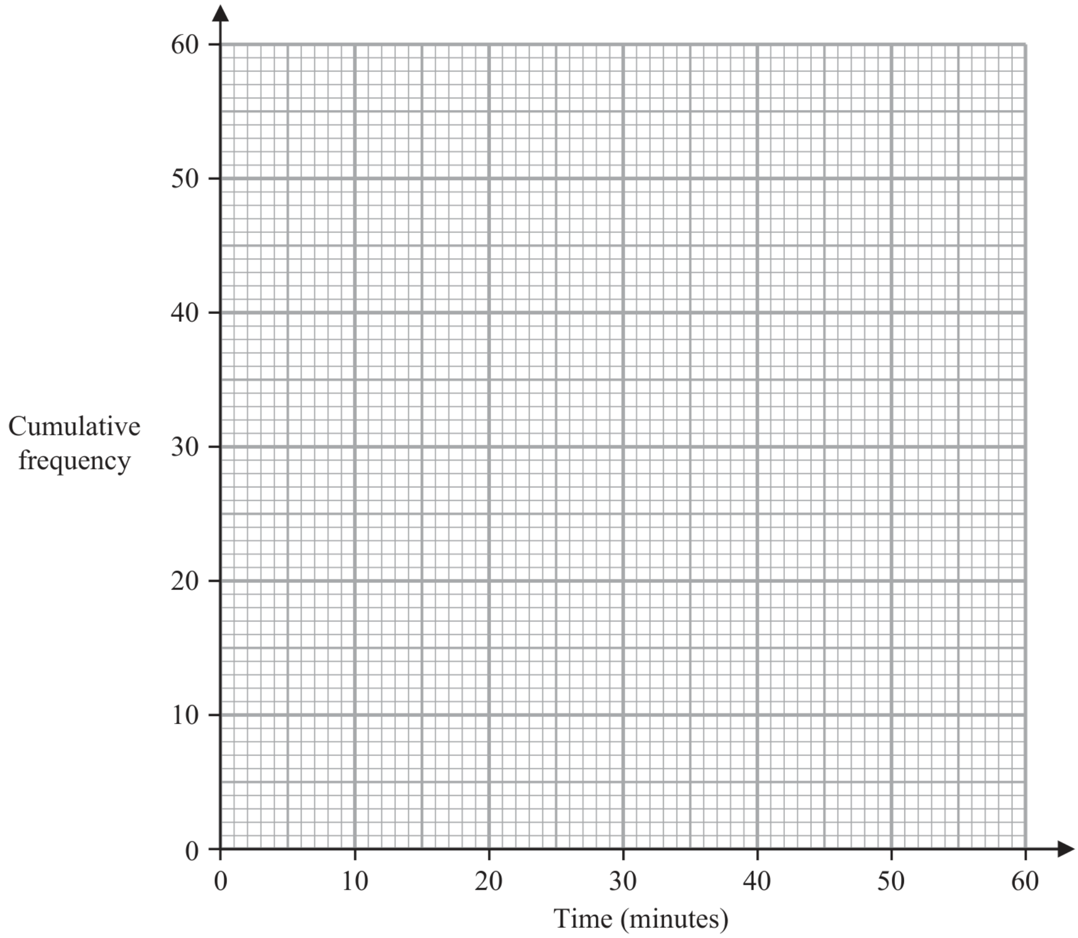
Time (t minutes)	**Cumulative frequency**
$0 < t \leqslant 10$	
$0 < t \leqslant 20$	
$0 < t \leqslant 30$	
$0 < t \leqslant 40$	
$0 < t \leqslant 50$	
$0 < t \leqslant 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued
4MA1-2H Q13

UNIT 2



(2)

(c) Use your graph to find an estimate for the median time.

.... minutes

(1)

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

....

(2)

(Total for Question 13 is 6 marks)

16 Make t the subject of the formula $c = \dfrac{t^2 + 3}{7 - 8t^2}$

....

(Total for Question 16 is 4 marks)

WORKING SPACE

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$\frac{dy}{dx} =$ ....

(2)

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

....

(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

19 Solve the inequality $4x^2 + 4x - 15 < 0$
Show clear algebraic working.

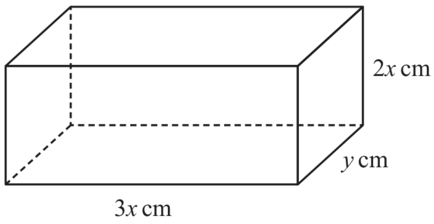
....

(Total for Question 19 is 3 marks)

WORKING SPACE

21 The diagram shows a cuboid.

Diagram **NOT** accurately drawn



The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is $1014\,\text{cm}^3$
The total surface area of the cuboid is $A\,\text{cm}^2$

Show that $A = 12x^2 + \dfrac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

WORKING SPACE

26 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

volume of solid **A** – volume of solid **B** $=$ 735 cm^3

find the volume of solid **A**

.... cm^3

(Total for Question 26 is 4 marks)

WORKING SPACE